

Name: \_\_\_\_\_

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# AREA AND PERIMETER, (AREA OF TRAPEZIUMS AND COMPOSITE SHAPES)

**LO: I can calculate the area of trapeziums and composite shapes by splitting into simpler shapes.**

## Trapezium and composite area

A **trapezium** has one pair of parallel sides (a and b). Its area is  $\frac{1}{2}(a + b) \times h$ .

**Composite shapes:** Split the shape into rectangles, triangles, and/or trapeziums. Find each area, then add (or subtract) them.

## Key facts

|   |                                      |                            |
|---|--------------------------------------|----------------------------|
| ● | Trapezium: $A = \frac{1}{2}(a + b)h$ | parallel sides a, b        |
| ● | Composite: split into known shapes   | then add areas             |
| ● | a = 5, b = 9, h = 4    A = 28        | $\frac{1}{2}(14) \times 4$ |
| ● | Trapezium: $A = a \times b \times h$ | wrong formula              |
| ● | Composite: just add all lengths      | that gives perimeter       |

## Key idea

Trapezium: average the parallel sides, then multiply by the height.

$$A = \left( \frac{1}{2}(a + b) \right) \times h$$

Average of parallel sides times height

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 - basic trapezium

A trapezium has parallel sides 6 cm and 10 cm, height 4 cm. Find the area.

$$A = \frac{1}{2}(a + b) \times h$$

$$A = \frac{1}{2}(6 + 10) \times 4 = \frac{1}{2} \times 16 \times 4$$

$$A = 32 \text{ cm}^2$$

$$= 32 \text{ cm}^2$$

Add the parallel sides, halve, then multiply by height.

### Example 2 - finding height

A trapezium has area  $30 \text{ cm}^2$ , parallel sides 4 cm and 8 cm. Find h.

$$30 = \frac{1}{2}(4 + 8) \times h$$

$$30 = 6h$$

$$h = 5 \text{ cm}$$

$$h = 5 \text{ cm}$$

Rearrange:  $h = A \div \frac{1}{2}(a + b)$ .

### Example 3 - L-shaped composite

An L-shape: outer  $8 \text{ cm} \times 6 \text{ cm}$ , cut-out  $3 \text{ cm} \times 4 \text{ cm}$ . Find area.

$$\text{Full rectangle: } 8 \times 6 = 48 \text{ cm}^2$$

$$\text{Cut-out: } 3 \times 4 = 12 \text{ cm}^2$$

$$\text{Area} = 48 - 12 = 36 \text{ cm}^2$$

$$= 36 \text{ cm}^2$$

Subtract the cut-out from the full rectangle.

### Example 4 - T-shaped composite

A T-shape: top rectangle  $10 \text{ cm} \times 2 \text{ cm}$ , stem  $3 \text{ cm} \times 6 \text{ cm}$ .

$$\text{Top: } 10 \times 2 = 20 \text{ cm}^2$$

$$\text{Stem: } 3 \times 6 = 18 \text{ cm}^2$$

$$\text{Total} = 20 + 18 = 38 \text{ cm}^2$$

$$= 38 \text{ cm}^2$$

Split into two rectangles and add.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Trapezium area

Find the area of each trapezium.

- a)  $a = 5 \text{ cm}$ ,  $b = 9 \text{ cm}$ ,  $h = 4 \text{ cm}$
- b)  $a = 3 \text{ cm}$ ,  $b = 7 \text{ cm}$ ,  $h = 6 \text{ cm}$
- c)  $a = 8 \text{ cm}$ ,  $b = 12 \text{ cm}$ ,  $h = 5 \text{ cm}$
- d)  $a = 4 \text{ cm}$ ,  $b = 10 \text{ cm}$ ,  $h = 3 \text{ cm}$

### 2. Finding missing measurements

Use the formula to find the missing value.

- a)  $A = 24$ ,  $a = 4$ ,  $b = 8$ . Find  $h$ .
- b)  $A = 35$ ,  $a = 3$ ,  $h = 5$ . Find  $b$ .
- c)  $A = 40$ ,  $b = 12$ ,  $h = 4$ . Find  $a$ .
- d)  $A = 27$ ,  $a = 5$ ,  $b = 4$ . Find  $h$ .

### 3. L-shapes and U-shapes

Find the area of each composite shape.

- a) L-shape:  $10 \times 8$ , cut-out  $4 \times 5$
- b) L-shape:  $12 \times 6$ , cut-out  $3 \times 3$
- c) U-shape:  $8 \times 6$ , cut-out  $4 \times 3$  (centre)
- d) L-shape:  $9 \times 7$ , cut-out  $5 \times 4$

### 4. T-shapes and plus signs

Split into rectangles, then add.

- a) T-shape: top  $8 \times 2$ , stem  $3 \times 5$
- b) T-shape: top  $12 \times 3$ , stem  $4 \times 6$
- c) Plus:  $10 \times 2$  and  $2 \times 10$  overlap  $2 \times 2$
- d) Arrow: rectangle  $6 \times 4$  + triangle  $b=6$ ,  $h=3$

### 5. Mixed composite and trapezium

Find each area.

- a) Trapezium:  $a = 6$ ,  $b = 14$ ,  $h = 8$
- b) Composite: rectangle  $5 \times 10$  + triangle  $b=5$ ,  $h=4$
- c) Trapezium:  $a = 7$ ,  $b = 11$ ,  $h = 6$
- d) L-shape:  $15 \times 10$ , cut-out  $8 \times 6$
- e) Pentagon (split): rectangle  $8 \times 5$  + triangle  $b=8$ ,  $h=3$

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Trapezium area =  $\frac{1}{2}(a + b) \times h$

☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b) Composite area: add all the sides together

☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c) Split composite shapes into rectangles and triangles

☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d) Trapezium area =  $a \times b \times h$

☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e) For an L-shape, subtract the cut-out from the full rectangle

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Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A trapezium has area  $60 \text{ cm}^2$ . One parallel side is **twice** the other. The height is **8 cm**. Find both parallel sides.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) A room is an L-shape with outer dimensions  $10 \text{ m} \times 8 \text{ m}$  and a  $4 \text{ m} \times 3 \text{ m}$  rectangular piece missing from one corner. Carpet costs **£12.50 per  $\text{m}^2$** . How much will it cost to carpet the room?

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

